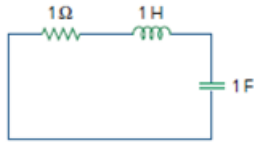


Problem

Refer to the series RLC circuit in Fig. 8.59. What kind of response will it produce?

- (a) overdamped
- (b) underdamped
- (c) critically damped
- (d) none of the above

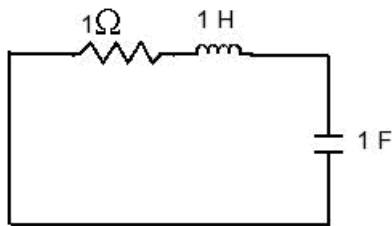
Fig. 8.59:



Step-by-step solution

Step 1 of 2

Given circuit diagram:



Step 2 of 2

From this circuit $\alpha = \frac{R}{2L}$

$$\alpha = \frac{1}{2 \times 1}$$

$$\alpha = \frac{1}{2}$$

$$\omega_o = \frac{1}{\sqrt{LC}}$$

$$\omega_o = \frac{1}{\sqrt{1 \times 1}}$$

$$\omega_o = 1$$

From this we can write $\alpha < \omega_o$, so the circuit is producing underdamped response.

Answer: (b)